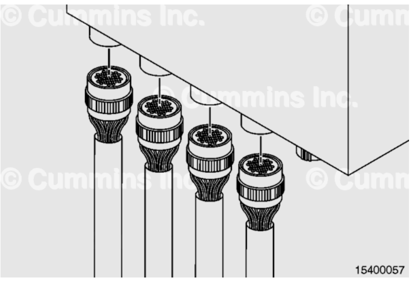
Trainings

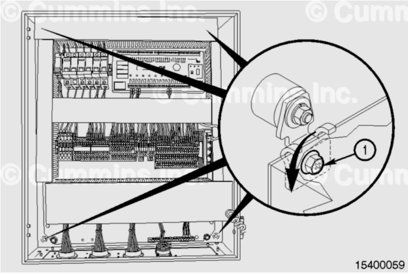
Remove

Customer Interface Box

Disconnect the C1, C2, C3, and C4 connectors at the bottom of the Customer Interface Box (CIB) panel. The data link has a protective cap and does **not** need to be removed unless the data link is in use.

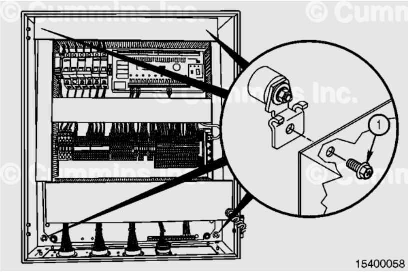


Loosen the mounting bolts (1) on the CIB panel mounting brackets.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



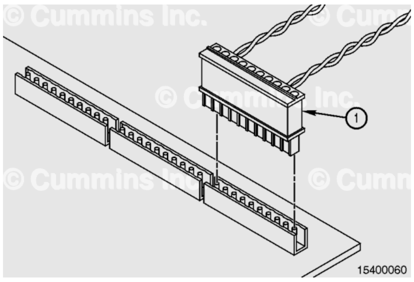
Support the CIB panel.

Remove the four mounting bolts (1).

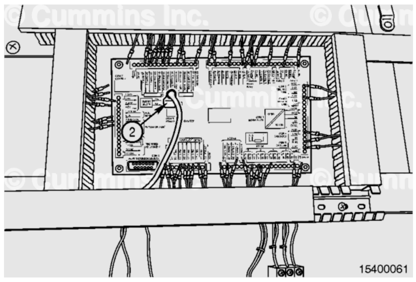
Diesel Control Unit

Make sure all wire identification is attached to the wires before removing the terminal strip plug connectors from the circuit board.

Disconnect the wires from the diesel control unit (DCU) by unplugging the terminal strips (1) from the circuit board.

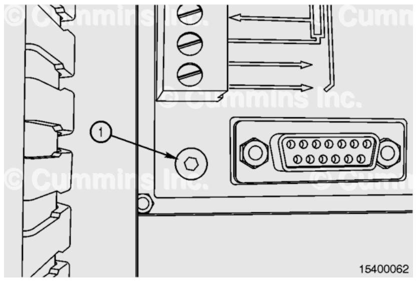


Disconnect the Ethernet Modbus™ connector (2) from the DCU.



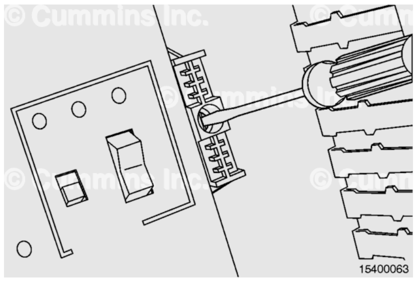
Remove the DCU mounting screws (1) from the front plate on the outside of the CIB.

Slide the DCU out of the front of the CIB panel.

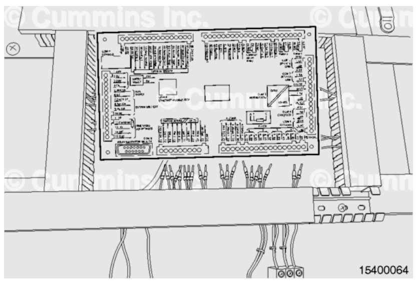


Loosen the spring clamps that hold the circuit board, to aid in removal.

Use a flat head screwdriver to loosen the screw in the clamp.



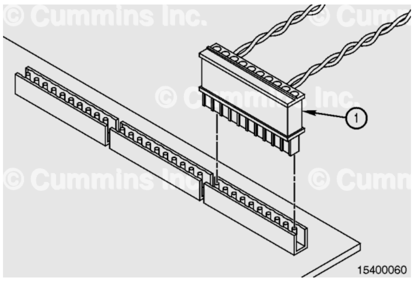
Remove the DCU circuit board from the CIB and rail by pressing down and lifting outward at the bottom of the circuit board.



Shutdown Unit

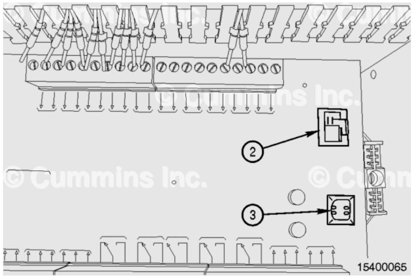
Make sure all wire identification is attached to the wires before removing the terminal strip plug connectors from the circuit board.

Disconnect the wires from the shutdown unit (SDU) by unplugging the terminal strips (1) from the circuit board.



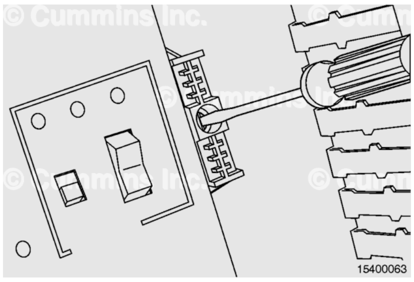
Disconnect the Com 3 configuration Ethernet connector (2) from the SDU.

Disconnect the Com 4 USB connector (3) from the SDU.

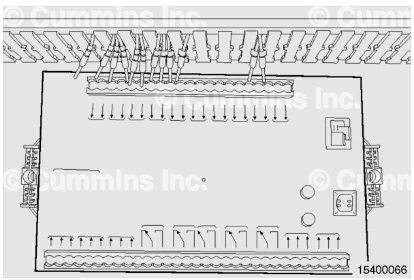


Loosen the spring clamps that hold the circuit board, to aid in removal.

Use a flat head screwdriver to loosen the screw in the clamp.



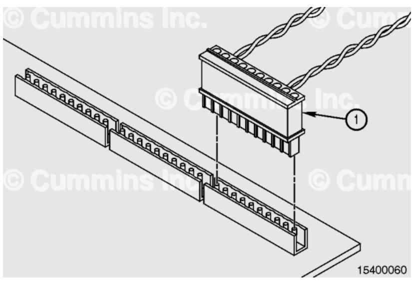
Remove the SDU circuit board from the CIB and rail by pressing down and lifting outward on the bottom of the circuit board.



Remote Input/Output Unit

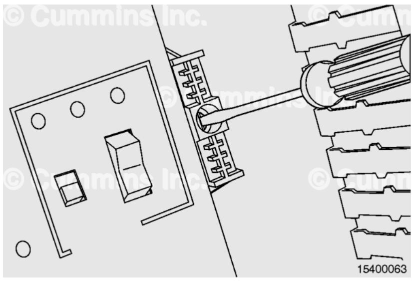
Make sure all wire identification is attached to the wires before removing the terminal strip plug connectors from the circuit board.

Disconnect the wires from the Remote Input/Output Unit (RIO) by unplugging the terminal strips (1) from the circuit board.

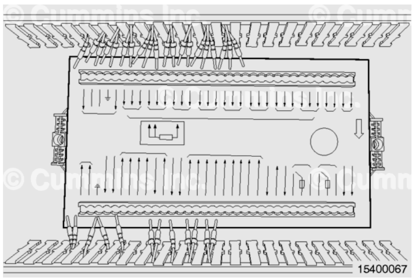


Loosen the spring clamps that hold the circuit board, to aid in removal.

Use a flat head screwdriver to loosen the screw in the clamp.



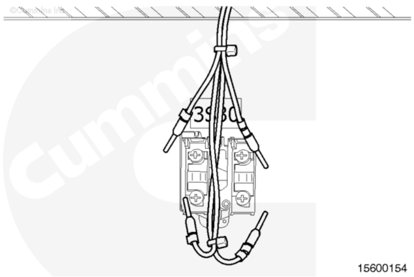
Remove the RIO circuit board from the CIB and rail by pressing down and pulling outward on the bottom of the circuit board.



Engine Stop Button

⚠ WARNING ⚠

To reduce the possibility of personal injury and equipment damage, shipboard tag out and lock out procedures must be followed.



Make a note of the connection points of all wires.

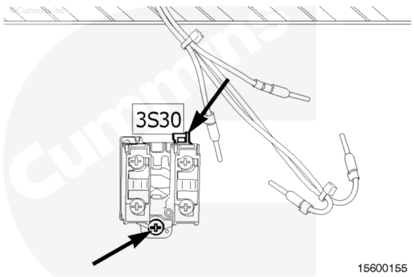
Loosen the wire set screws and remove the wires from the switch.

Note : Panels for the offshore drill rig application will have a third contact with corresponding wires in this location.

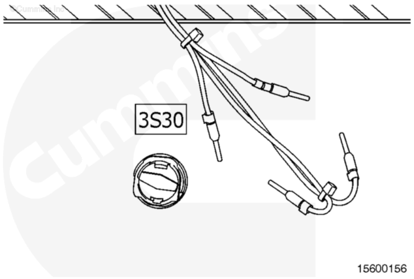
Loosen the bottom set screw.

Lift the upper lever.

Pull the rear switch assembly off of the button.



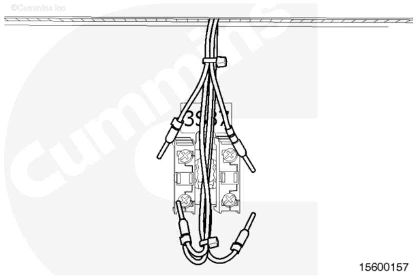
Slide the engine stop button out of the panel door.



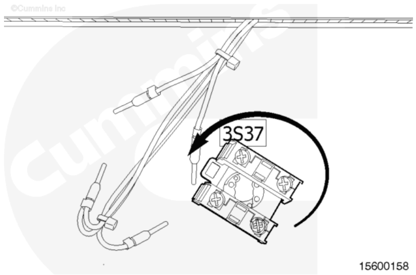
Power Switch

Make a note of the connection points of all wires.

Loosen the wire set screws and remove the wires from the switch.

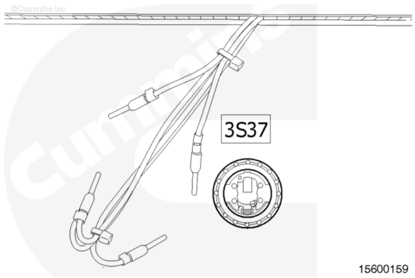


Remove the switch assembly by turning the switch assembly **counterclockwise** and pulling outward.



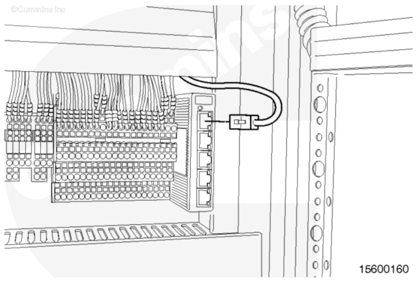
Turn the button mounting nut **counterclockwise** and remove the nut.

Slide the button out of the panel.



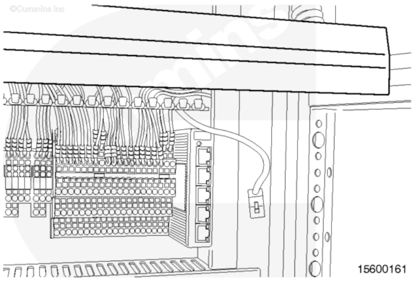
Ethernet Switch

Disconnect the Ethernet cable from the Ethernet switch.



Remove the cable tray cover.

Disconnect the Ethernet switch power connector.

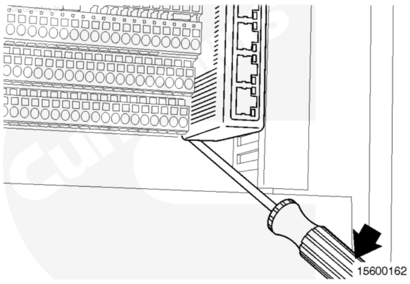


Loosen the terminal strip securing mechanism set screw.

Slide the terminal strip securing mechanism to the right as much as possible.

Place a flat head screwdriver in the bottom tab of the Ethernet switch.

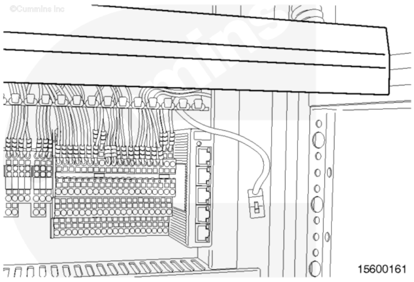
To remove the Ethernet switch, apply downward pressure on the screwdriver handle while lifting up on the Ethernet switch.



Terminal Strip

Remove the cable tray cover.

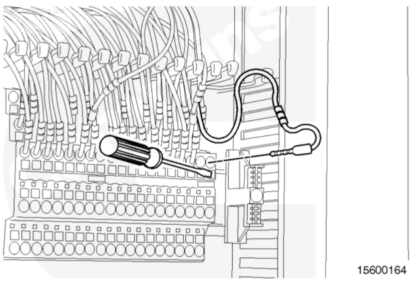
Remove the Ethernet switch.



Make a note of the connection points of all wires.

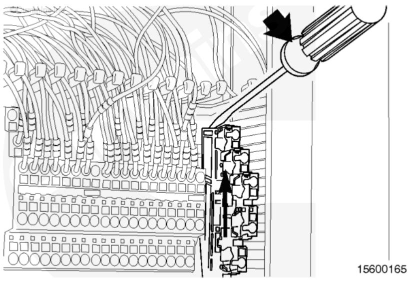
Place a flat head screwdriver in the slot below the wire to be removed.

Apply downward pressure on the screwdriver handle and remove the wire(s).



Place a flat head screwdriver in the upper slot of the connector.

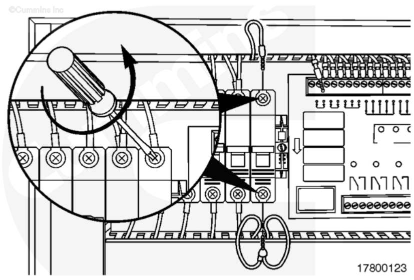
To remove the terminal strip connector, apply downward pressure on the screwdriver handle while lifting up on the connector.



Circuit Breaker

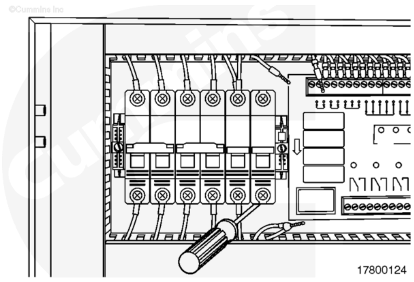
Make a note of the wire location of the wires on the circuit breaker.

Loosen the set screws and remove the wires.



Place a flat head screwdriver in the lower release rectangle and move the handle upward to release the circuit breaker from the track.

Remove the circuit breaker.

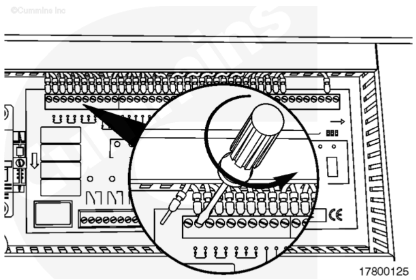


Logic Unit

Note all locations of the wires on the customer interface box logic unit.

Loosen the set screws on the wires.

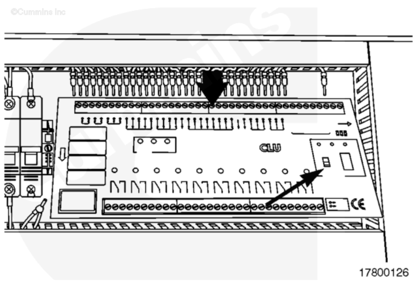
Remove the wires from the customer interface box logic unit.



Press down on the top of the customer interface box logic unit.

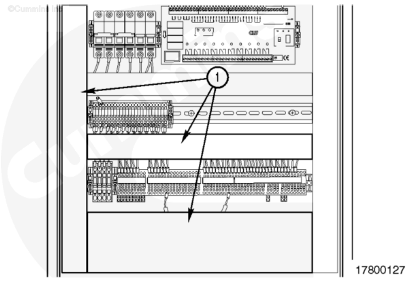
At the end of travel, pull outward on the bottom side.

Remove the customer interface box logic unit.



Conductor

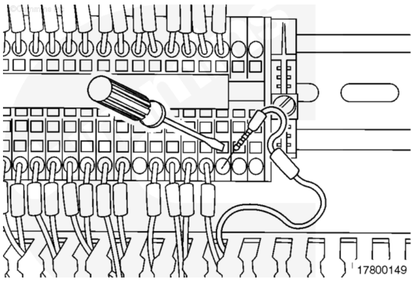
Remove the associated cable tray covers (1).



Note the location of the wire to be removed.

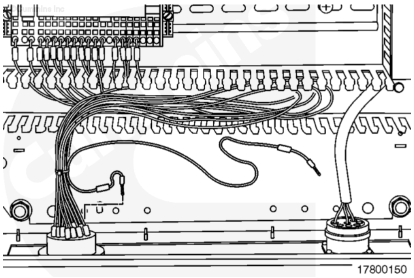
Place the screwdriver in the square slot on the terminal strip.

Move the handle in an downward motion and remove the wire.



Remove the wire from the cable tray.

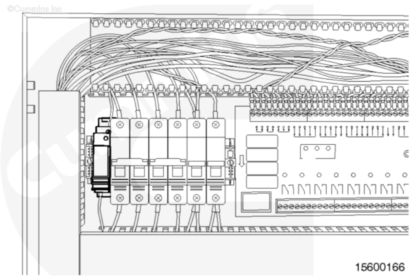
Remove the wire from the other connection point.



Engine Protection Override Relay

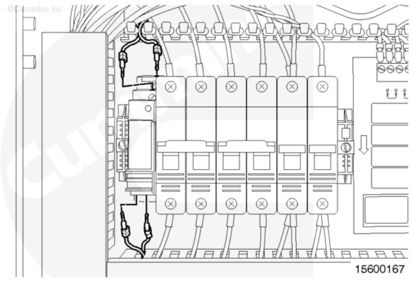
- Solid state relay
- Mounted in a block.

Remove the cable tray cover.



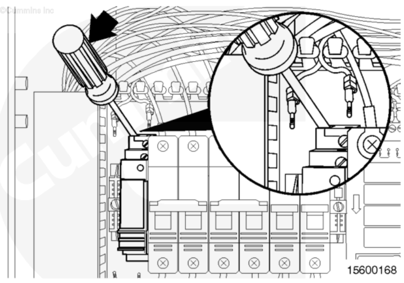
Make a note of the connection points of all wires.

Loosen the engine protection override relay terminal screws.



Place a flat head screwdriver in the tab on the engine protection override relay block.

Apply downward pressure on the screwdriver handle while pushing up on the engine protection override relay block and remove the block and relay assembly.

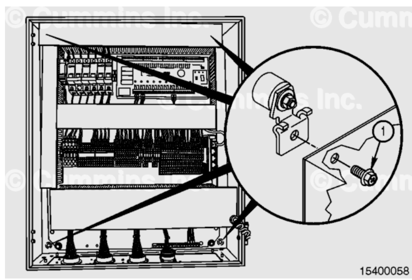


Install

Customer Interface Box

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

An isolator must be used when mounting a CIB panel. Vibration can damage the panel and internal components.

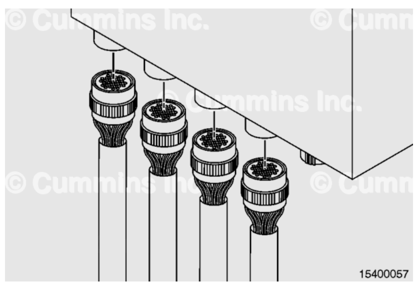
Support the CIB panel.

Install the four mounting bolts (1) through the panel and into the mounting brackets.

Tighten the mounting bolts.

Torque Value: 64 n•m [47 ft-lb]

Connect the C1, C2, C3, and C4 connectors at the bottom of the Customer Interface Box (CIB) panel. Connect the data link if it was previously disconnected.

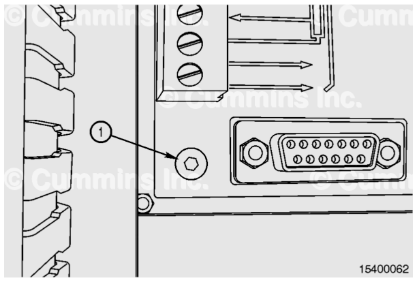


Diesel Control Unit

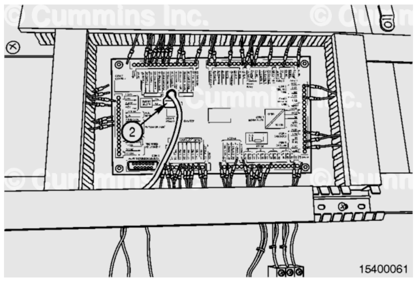
Install the diesel control unit (DCU) through the front of the CIB door and install the 4 mounting screws (1).

Tighten the screws.

Torque Value: 0.4 n•m [3.6 in-lb]

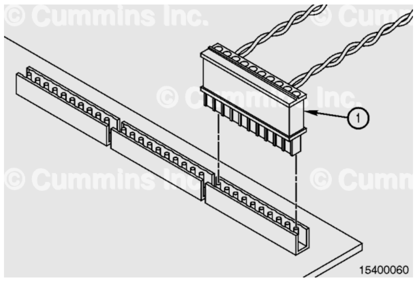


Connect the Ethernet Modbus™ connector (2) to the DCU.



Make sure all wire identification on the terminal strip connector matches the DCU.

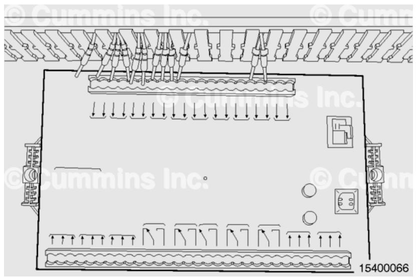
Connect the terminal strip connector (1) to the circuit board.



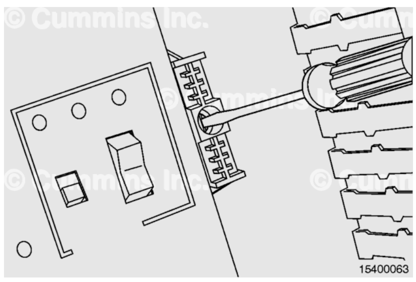
Shutdown Unit

Install the SDU circuit board by latching the unit on the top of the DIN rail and pressing the unit firmly into place.

The unit will make an audible noise when it seats into position.

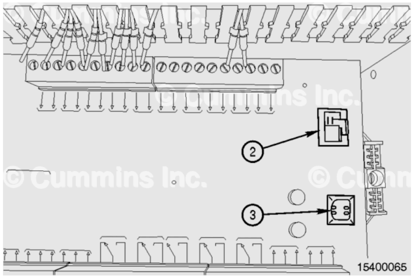


If loosened for removal, tighten the spring clamps with a flat head screwdriver.



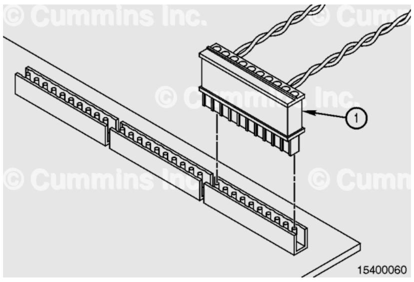
Connect the Com 3 configuration Ethernet connector (2) to the SDU.

Connect the Com 4 USB connector (3) to the SDU.



Make sure all wire identification matches the circuit board.

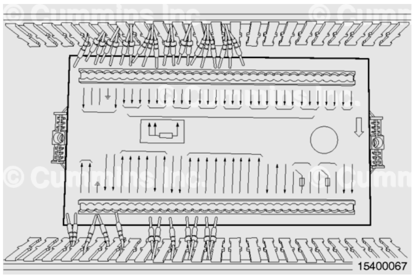
Connect the terminal strip connector (1) to the circuit board.



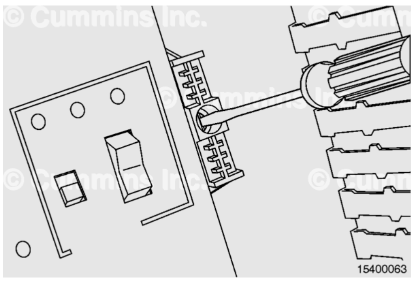
Remote Input/Output Unit

Install the RIO circuit board by latching the unit on the top of the DIN rail and pressing the unit firmly into place.

The unit will make an audible noise when it seats into position.

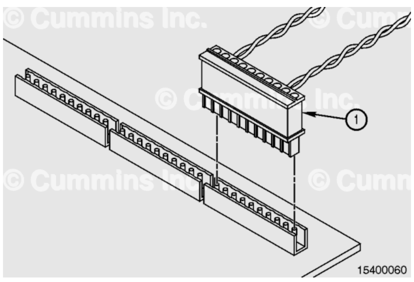


If loosened for removal, tighten the spring clamps with a flat head screwdriver.



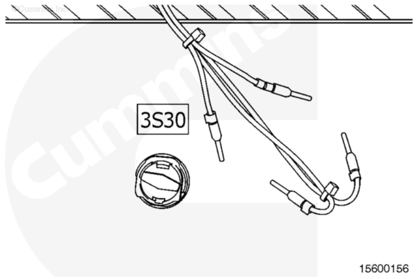
Make sure all wire identification matches the circuit board.

Connect the terminal strip connector (1) to the circuit board.



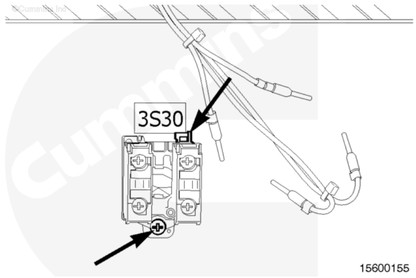
Engine Stop Button

Slide the engine stop button into the panel door.



Lift the lever on the switch assembly and place onto the button.

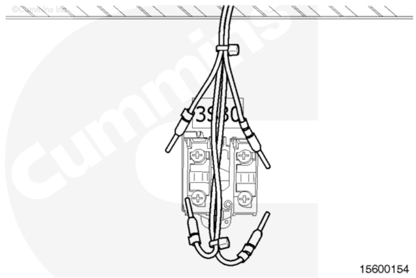
Tighten the bottom set screw.



Install the wires into the locations they were removed from.

Tighten the set screws.

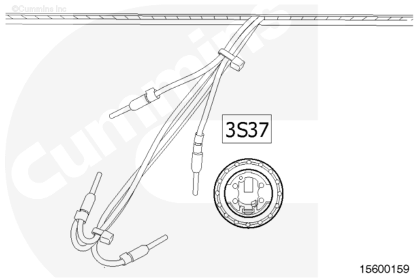
Note : Panels for the offshore drill rig application will have a third contact and corresponding wires.



Power Switch

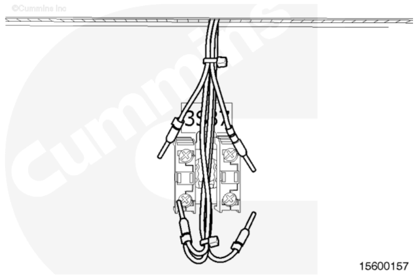
Slide the button into the panel.

Install the button nut and tighten.



Place the switch assembly onto the button assembly and push it until it snaps onto the button assembly.

Install the wires into the locations they were removed from and tighten the set screws.

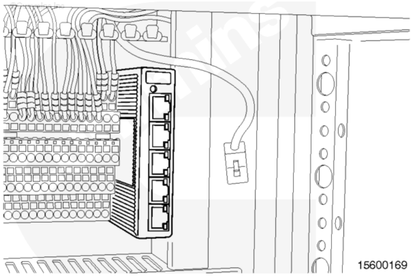


Ethernet Switch

Place the Ethernet switch on the din rail connector with the top of the switch connecting assembly on the top of the din rail.

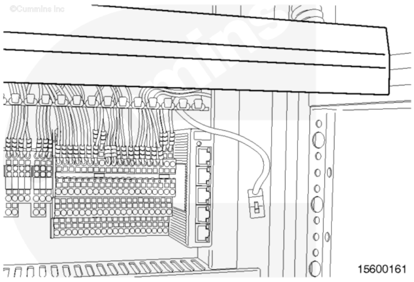
Push the bottom of the switch downwards until it snap onto the track.

Slide the terminal strip securing mechanism to the left and tighten the set screw.

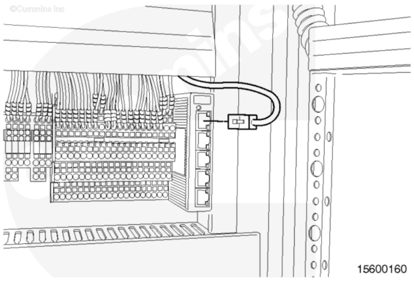


Connect the Ethernet switch power connector.

Install the cable tray cover.



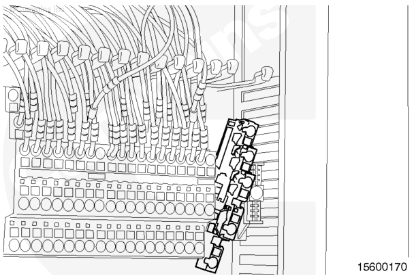
Connect the Ethernet cable to the Ethernet switch.



Terminal Strip

Position the upper leg of the terminal strip onto the track.

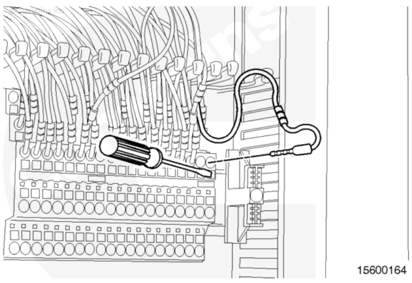
Press the lower portion of the terminal strip until it snaps onto the track.



Place a flat head screwdriver in the slot below the wire to be installed.

Apply downward pressure on the screwdriver handle and install the wire(s).

Install the cable tray cover.

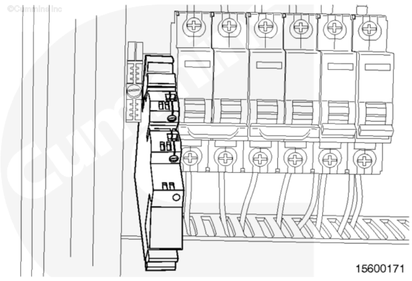


Engine Protection Override Relay

- Solid state relay
- Mounted in a block.

Place the lower leg of the engine protection override relay block on the track.

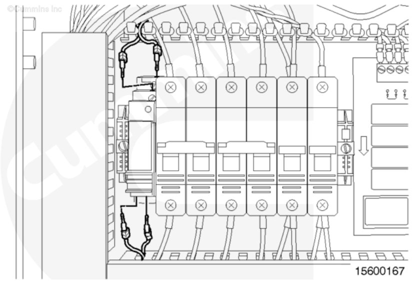
Press the upper portion of the block until it snaps into place.



Install the wires into the engine protection override relay block.

Tighten the setscrews.

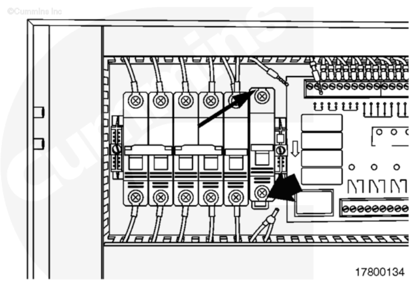
Install the cable tray cover.



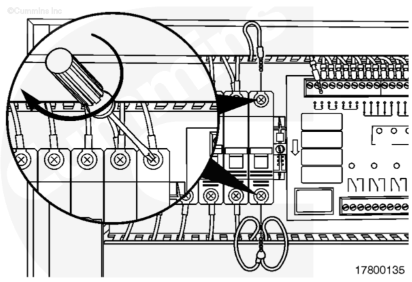
Circuit Breaker

Place the upper connecting tab on the track.

Push the lower portion of the circuit breaker until it locks onto the track.



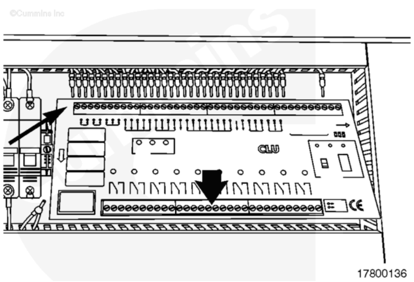
Install the wires into the noted locations and tighten the set screws.



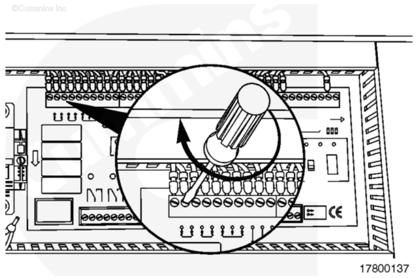
Logic Unit

Place the upper portion of the logic unit on the track.

Push the lower portion of the logic unit until it locks onto the track.

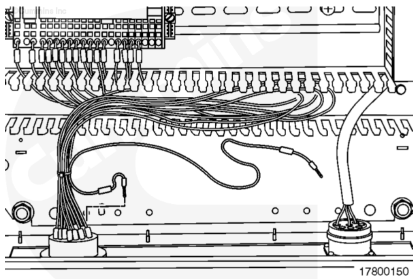


Install the wires into the noted locations and tighten the set screws.



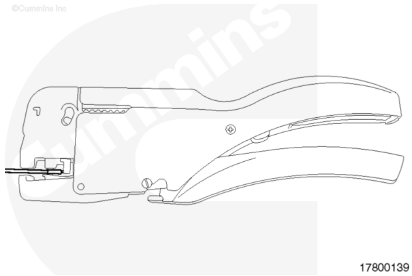
Conductor

Route the wire through the cable tray to the associated connector.



⚠ WARNING ⚠

To reduce the possibility of personal injury and equipment damage, the wire must have the correct ferrule installed with the stripping and crimp tool.

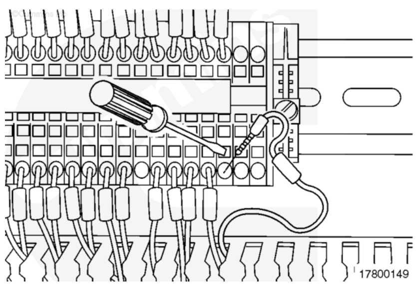


Strip the wire with the stripping and crimp tool, Part Number 4918722.

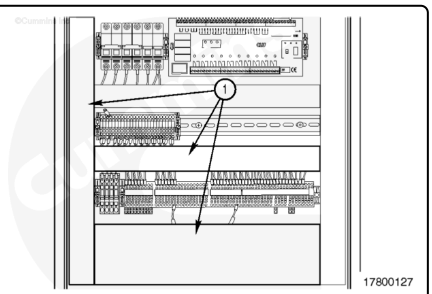
Crimp the new ferrule onto the end of the wire.

Place the screwdriver in the square slot on the terminal strip.

Move the handle in an downward motion and insert the wire.



Install the cable tray covers (1).



Last Modified: 25-Jan-2010